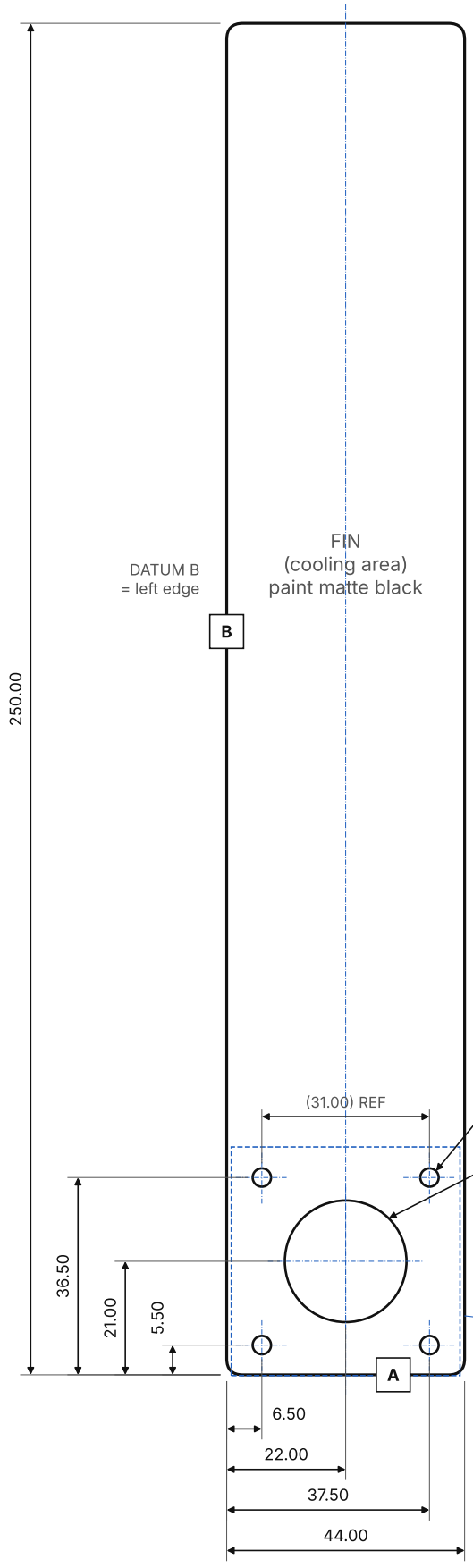
FRONT VIEW (PETG tab side)



4X Ø3.40 THRU (B-E)
M3 clearance, NEMA 17 pattern

Ø22.50 THRU (A)
motor boss Ø22.0 + clearance

motor contact face 42.3 □:
leave BARE, thermal paste

EDGE VIEW



NOTES

1. MATERIAL: 6061-T6 OR 5052-H32 ALUMINIUM, 1/8" (3.175) PLATE.
2. QTY 4 (ONE PER WINCH STATION).
3. ALL HOLES THRU. HOLE POSITIONS ±0.10 RELATIVE TO EACH OTHER; DRILL ALL 5 IN ONE SETUP FROM DATUMS A AND B. DATUM A = BOTTOM EDGE (SITS ON DECK), B = LEFT EDGE.
4. BLANK 44.0 x 250.0 ±0.5 (SHEARED). BREAK ALL EDGES; CORNER R3 OPTIONAL.
5. DEBURR BOTH FACES. MOTOR FACE FLAT, NO RAISED BURRS.
6. FINISH: MATTE BLACK PAINT EXCEPT THE 42.3 SQ. MOTOR CONTACT AREA (BOTH FACES MAY BE PAINTED ABOVE IT).
7. ASSEMBLY: MOTOR | FIN | 5 MM PETG TAB, M3 x 12 BOLTS, THERMAL PASTE ON MOTOR FACE. MOTOR + SPOOL SHIFT 3.2 ALONG THE SHAFT.
8. MASS ≈ 91 g EACH.

PART	MOTOR HEAT FIN (motor_heat_plate)
PROJECT	RoomCleaner winch station, gen 2
MATERIAL	AL 6061-T6 / 5052-H32, 3.175 PLATE
UNITS	mm
SCALE	NOT TO SCALE (0.8x) - use dimensions
GENERAL TOL.	±0.5 unless noted
REV	B 2026-10-19
SOURCE	cad/parts/motor_heat_plate.py
SHEET	1 OF 2

DIMENSIONS GOVERN - DO NOT SCALE THIS SHEET.
1:1 hole template on sheet 2.

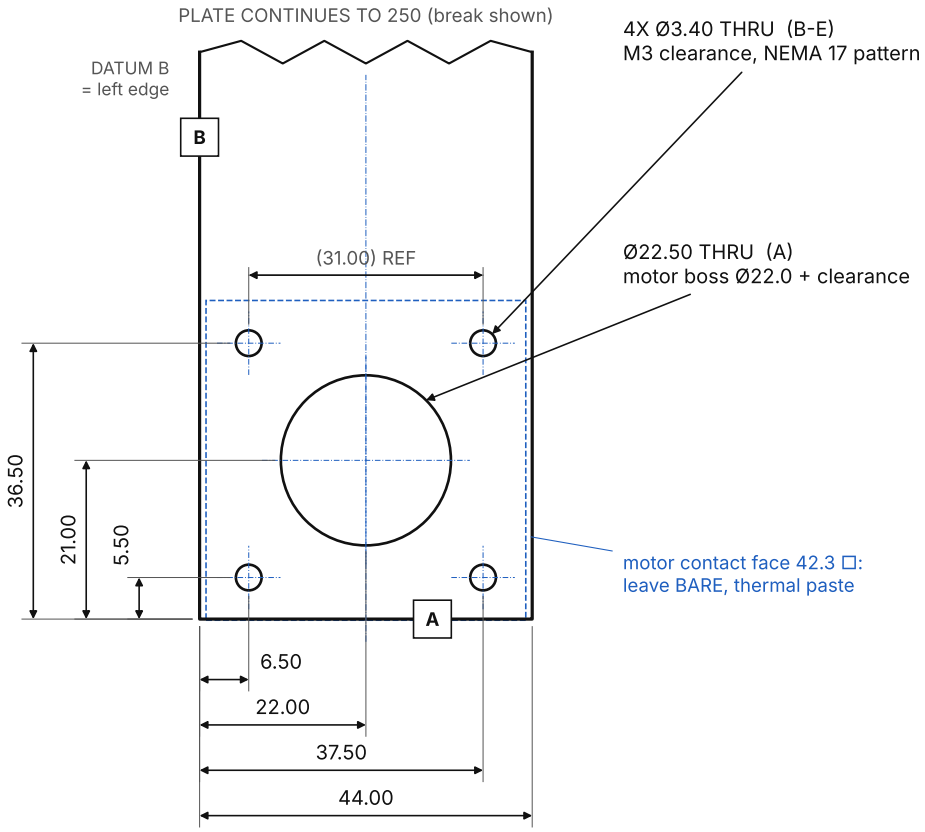
SHEET 2 - TRUE 1:1 HOLE TEMPLATE

PRINT AT 100 % / "ACTUAL SIZE". DO NOT "FIT TO PAGE".

Check the 100 mm bar below with a ruler before using.

Use: cut along the plate outline, align the bottom edge with datum A and the left edge with datum B, tape down, then centre-punch each crosshair. Machine to the dimensions; the template is a check / punch guide, not a gauge.

HOLE-END TEMPLATE 1:1



SCALE CHECK: this bar must measure 100 mm

PART	MOTOR HEAT FIN (motor_heat_plate)
PROJECT	RoomCleaner winch station, gen 2
MATERIAL	AL 6061-T6 / 5052-H32, 3.175 PLATE
UNITS	mm
SCALE	1:1 (print 100 %)
GENERAL TOL.	±0.5 unless noted
REV	B 2026-10-19
SOURCE	cad/parts/motor_heat_plate.py
SHEET	2 OF 2